```c
#include <stdio.h>

int main() {
    int num_blocks, num_processes;
    printf("Enter number of memory blocks: ");
    scanf("%d", &num_blocks);

    int blocks[num_blocks];
    printf("Enter sizes of %d memory blocks:\n", num_blocks);
    for (int i = 0; i < num_blocks; i++)
        scanf("%d", &blocks[i]);

    printf("Enter number of processes: ");
    scanf("%d", &num_processes);

    int processes[num_processes];
    printf("Enter sizes of %d processes:\n", num_processes);
    for (int i = 0; i < num_processes; i++)
        scanf("%d", &processes[i]);

    int temp[num_blocks];
    int allocation[num_processes];

    for (int i = 0; i < num_blocks; i++) temp[i] = blocks[i];
    for (int i = 0; i < num_processes; i++) allocation[i] = -1;

    printf("\n--- First Fit ---\n");
    printf("Process No. Process Size    Block No.\n");
```

```
for (int i = 0; i < num_processes; i++) {
    for (int j = 0; j < num_blocks; j++) {
        if (temp[j] >= processes[i]) {
            allocation[i] = j + 1;
            temp[j] -= processes[i];
            break;
        }
    }
    if (allocation[i] != -1)
        printf("%d\t\t%d\t\t%d\n", i + 1, processes[i], allocation[i]);
    else
        printf("%d\t\t%d\t\tNot Allocated\n", i + 1, processes[i]);
}
for (int i = 0; i < num_blocks; i++) temp[i] = blocks[i];
for (int i = 0; i < num_processes; i++) allocation[i] = -1;

printf("\n--- Best Fit ---\n");
printf("Process No. Process Size    Block No.\n");
for (int i = 0; i < num_processes; i++) {
    int best = -1;
    for (int j = 0; j < num_blocks; j++) {
        if (temp[j] >= processes[i]) {
            if (best == -1 || temp[j] < temp[best])
                best = j;
        }
    }
    if (best != -1) {
        allocation[i] = best + 1;
```

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            temp[best] -= processes[i];
            printf("%d\t\t%d\t\t%d\n", i + 1, processes[i], allocation[i]);
        } else {
            printf("%d\t\t%d\t\tNot Allocated\n", i + 1, processes[i]);
        }
    }
    for (int i = 0; i < num_blocks; i++) temp[i] = blocks[i];
    for (int i = 0; i < num_processes; i++) allocation[i] = -1;
    printf("\n--- Worst Fit ---\n");
    printf("Process No. Process Size    Block No.\n");
    for (int i = 0; i < num_processes; i++) {
        int worst = -1;
        for (int j = 0; j < num_blocks; j++) {
            if (temp[j] >= processes[i]) {
                if (worst == -1 || temp[j] > temp[worst])
                    worst = j;
            }
        }
        if (worst != -1) {
            allocation[i] = worst + 1;
            temp[worst] -= processes[i];
            printf("%d\t\t%d\t\t%d\n", i + 1, processes[i], allocation[i]);
        } else {
            printf("%d\t\t%d\t\tNot Allocated\n", i + 1, processes[i]);
        }
    }
    return 0;
}
```

OUTPUT :

```
Enter number of memory blocks: 5
Enter sizes of 5 memory blocks:
100 500 200 300 600
Enter number of processes: 4
Enter sizes of 4 processes:
212 417 112 426

--- First Fit ---
Process No. Process Size    Block No.
1               212             2
2               417             5
3               112             2
4               426             Not Allocated

--- Best Fit ---
Process No. Process Size    Block No.
1               212             4
2               417             2
3               112             3
4               426             5

--- Worst Fit ---
Process No. Process Size    Block No.
1               212             5
2               417             2
3               112             5
4               426             Not Allocated
```